

 OR-AS Operations Research Applications and Solutions	Case Name: New product development	Sector	Engineering	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
Submitted by	Rojin & Forough		One of nine std. scenarios	
Date	2024	Project Control	User defined distributions	
File Name	C2024-04 New product development		Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

The development and implementation of a new product, including design, testing, and production preparation activities.

The project consists of activity, resource, and cost data obtained directly from the actual project execution.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	37	Serial/Parallel (SP)	58%
Planned Duration (PD)	84d	Activity Distribution (AD)	57%
Budget At Completion (BAC)	32,481€	Length of Arcs (LA)	10%
Renewable Resources	-	Topological Float (TF)	8%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	11	14	2.43
CRI-rho	21	19	0.99
CRI-tau	27	37	1.45

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	54	37	0.00
SI	60	38	-0.58
SSI	9	13	2.71
CRI-r	11	12	2.03
CRI-rho	20	18	0.96
CRI-tau	27	37	1.48

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	6.83	-4.08
PV - SPI	7.08	1.72
PV - SCI	7.36	2.04
ED - 1	6.19	-3.15
ED - SPI	7.08	1.72
ED - SCI	7.25	1.76
ES - 1	6.60	-4.63
ES - SPI(t)	5.18	-1.47
ES - SCI(t)	5.34	-1.41

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	0.76	-0.14
CPI	0.85	-0.22
SPI	5.06	-0.14
SPI(t)	3.03	5.06
SCI	5.14	3.02
SCI(t)	3.16	5.11
0.8 CPI + 0.2 SPI	1.27	3.08
0.8 CPI + 0.2 SPI(t)	0.92	1.04

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over **2** tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	1060.64	1709.69	35.10	1.06	1.16	1.09	0.98
std dev	1272.78	2144.15	41.09	0.09	0.24	0.08	0.02
final	2643.00	0.00	-32.00	1.09	1.00	0.96	1.00

2.3.3.2. Time forecasting

PD	84d	Real Duration	239d	Late/early	184.52%
----	-----	---------------	------	------------	---------

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	N/A	N/A	N/A	N/A
PV - SPI	N/A	N/A	N/A	N/A
PV - SCI	N/A	N/A	N/A	N/A
ED - 1	N/A	N/A	N/A	N/A
ED - SPI	N/A	N/A	N/A	N/A
ED - SCI	N/A	N/A	N/A	N/A
ES - 1	N/A	N/A	N/A	N/A
ES - SPI(t)	N/A	N/A	N/A	N/A
ES - SCI(t)	N/A	N/A	N/A	N/A

2.3.3.3. Cost forecasting

BAC	32,481.00€	Real Cost	35,606.00 €	Over/under budget	9.62%
-----	------------	-----------	-------------	-------------------	-------

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	N/A	N/A	N/A	N/A
CPI	N/A	N/A	N/A	N/A
SPI	N/A	N/A	N/A	N/A
SPI(t)	N/A	N/A	N/A	N/A
SCI	N/A	N/A	N/A	N/A
SCI(t)	N/A	N/A	N/A	N/A
0.8 CPI + 0.2 SPI	N/A	N/A	N/A	N/A
0.8 CPI + 0.2 SPI(t)	N/A	N/A	N/A	N/A